

Multi-Defect Fusion Test

PAUT Explainable-AI Digital Twin — weld defect inspection from Total Focusing Method (TFM) images.

Generated by `scripts/multidefect\_fusion.py`. Every training image is single-type (one weld = one defect class), so a composite holding BOTH porosity and slag is out-of-distribution — this checks whether detection generalises to mixed defects.

Fused 8 porosity+slag pairs (test split) by amplitude max-blend.

Results

- **Both types detected in 88%** of composites.
- Porosity detected in 88%, slag detected in 100%.
- Mean per-class Dice on composites: porosity 0.590, slag 0.609.

Pair	porosity	slag	both	Dice por	Dice slag
porosity__G110_1+slag__GS105	Y	Y	Y	0.63	0.46
porosity__G110_2+slag__GS105_1	Y	Y	Y	0.73	0.70
porosity__G110_3+slag__GS105_2	Y	Y	Y	0.40	0.30
porosity__G120_1+slag__GS105_3	Y	Y	Y	0.99	0.92
porosity__G120_2+slag__GS117	-	Y	-	0.00	0.55
porosity__G120_3+slag__GS117_1	Y	Y	Y	0.00	0.00
porosity__G141_1+slag__GS117_2	Y	Y	Y	0.97	0.98
porosity__G141_2+slag__GS117_3	Y	Y	Y	0.99	0.98

Interpretation

- **the segmentation head generalises well to mixed defects — it localises both flaw types in one image despite only ever training on single-type images.**
- The miss is asymmetric: **porosity** is detected less often (porosity 88% vs slag 100%). This matches its weaker stand-alone score (porosity Dice ~0.45 vs slag ~0.68): when the fainter porosity is max-blended with the stronger slag, the dominant signal suppresses it.
- Detection is per-pixel, so the head *can* light up both class channels when both signatures are clearly present (see the both-detected cases) — the limit is signal strength of the weaker flaw, not the architecture.
- **Mitigation:** include a few multi-defect composites in training, or detect each class at its own threshold. Gallery: data/processed/fusion/fusion_gallery.png.